

Trading Crypto Funding Rates — Part 1

Do cheaper-funding coins earn better returns?



DIMA

19 HOURS AGO



5



1

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In the [breakout study](#), I tested price trends. This time I want to test funding.

The idea is simple: buy the coins where funding is cheaper, and short the ones where it is more expensive. If that difference lasts, the payments should help us. The problem is that we still own the price risk.

Does buying coins with relatively cheap funding and shorting coins with relatively expensive funding make money after price moves and trading costs?

As with breakout, I want to show the process, including the parts that did not work. In Part 1 I test whether funding contains useful information, then try the simplest trade. In [Part 2](#) I test whether trading less and changing position sizes can help us keep more of the return.

What is funding?

Perpetual futures have no expiry. Funding transfers payments between longs and shorts to help keep the contract near its spot market.

- Positive funding: longs pay shorts.
- Negative funding: shorts pay longs.
- Payments apply to the position's notional value.

Eg a \$10,000 short receives \$10 at a +0.10% settlement. If price rises 2%, it loses \$200 on price. We collected funding and still lost \$190 before fees.

I add the recorded funding rates within each UTC day, average the preceding completed days, then multiply by 365. That gives an annualised rate for comparison. Payments rather than assuming three

Cheap means relative; its rate is lower than the funding and only slightly

For example, +10% funding at +10%; it

The timing is simple: the price move and wait an extra day before

Universe

The study uses Binance data from 2023–2025. I include retired contracts rather than just today's survivors, although the archive does not cover every historical listing. Checks against raw data agree where they overlap; I cannot guarantee every settlement is recorded.

These are historical tests, not predictions of live performance.

The first test uses every contract in the archive. Then I compare larger and smaller coins using market caps known before each month. Our source covers the global Top 200 and gives complete mapped Top 20, Top 50 and Top 100 groups; smaller Binance contracts can be missing.

The historical universe helps avoid selecting only the coins that survived. It does not remove the archive's coverage limits.



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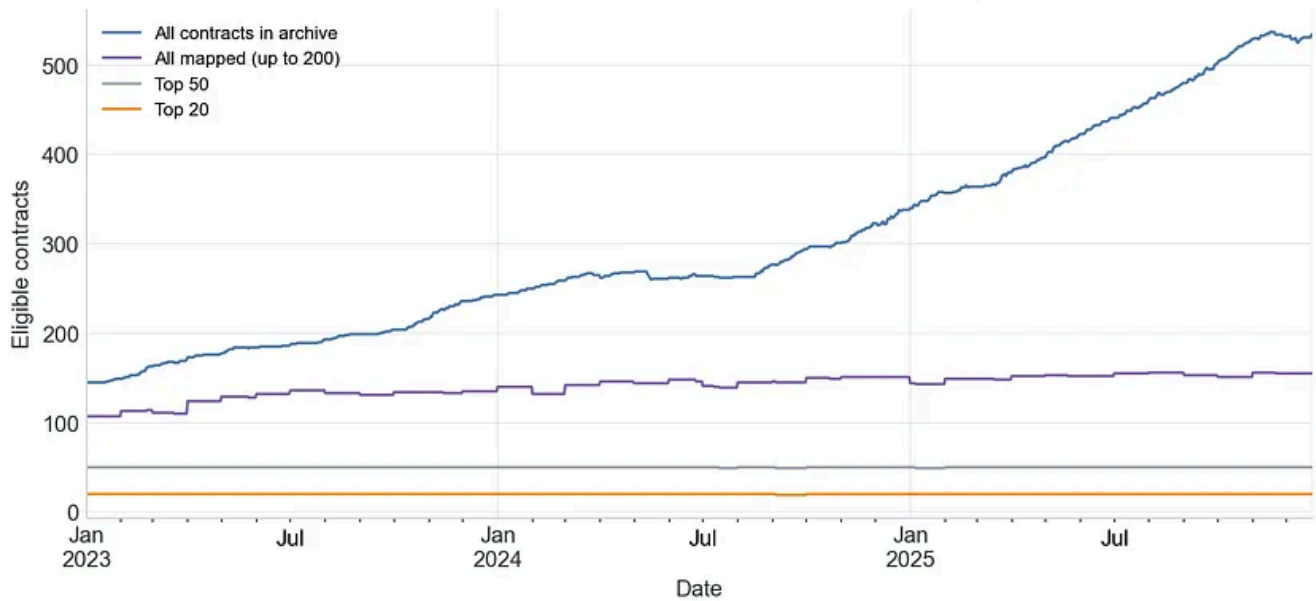
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How much of Binance can we actually study?



2023–2025 · historical contracts, not current survivors · market-cap snapshot known before each month · source covers the global Top 200 · counts before future-outcome checks.

How sticky is funding?

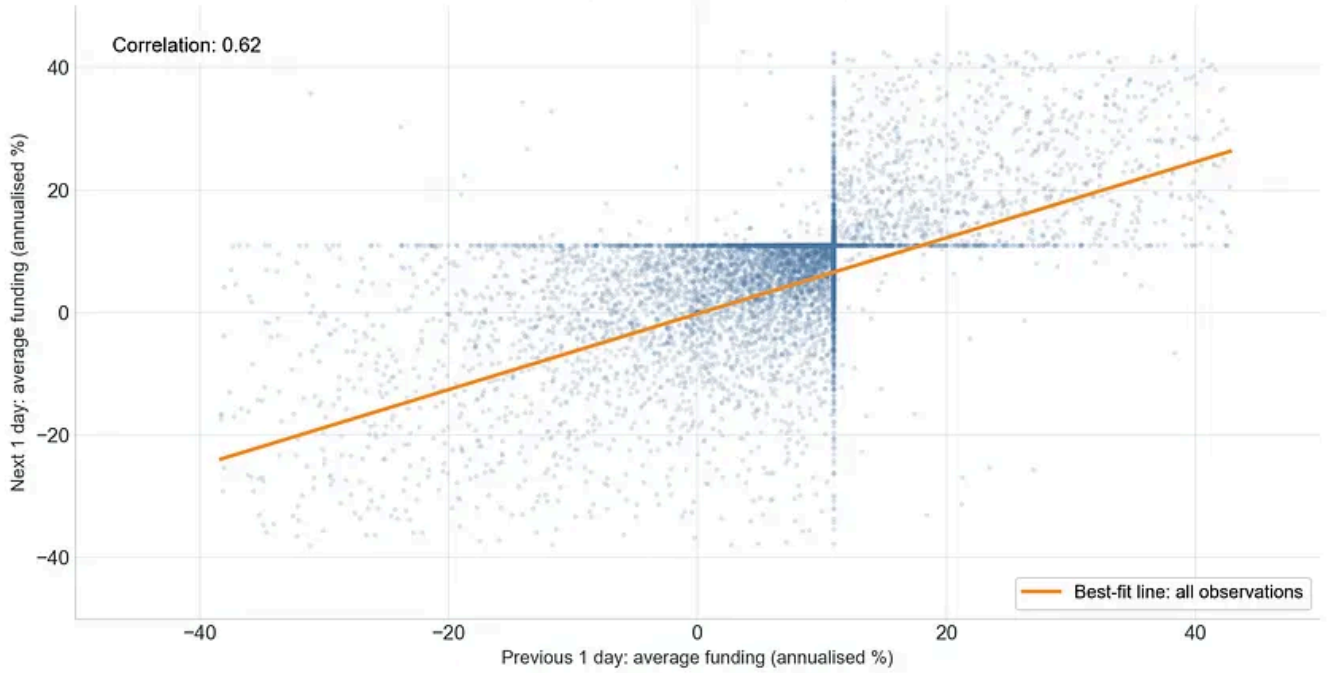
First, a simple question: if funding is high today, does it tend to stay high tomorrow?

If the payment disappears before I can collect it, yesterday's attractive rate is not much use. Persistence is the first thing to check, not the final reason to trade.

I compare today's funding with tomorrow's. Then I repeat the test using the average over the previous 3 days versus the next 3, and the previous 7 days versus the next 7. Each earlier window ends before the next begins.

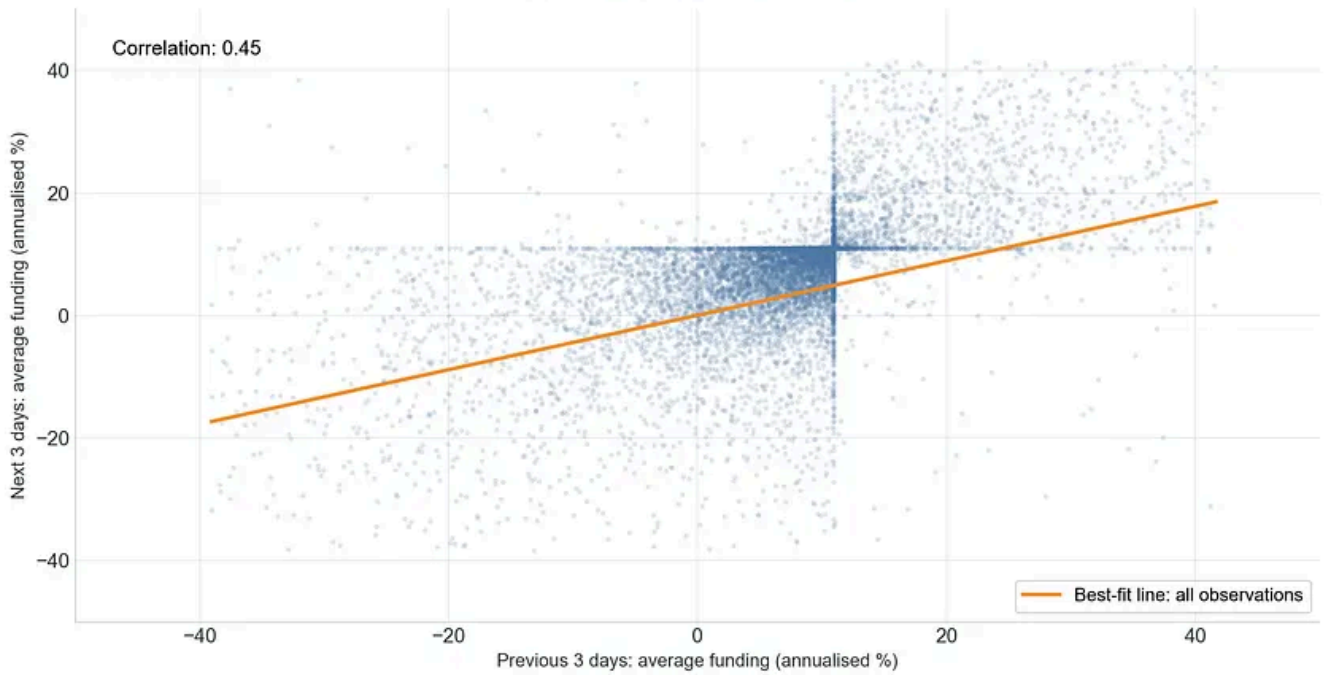
Each dot pairs a coin's earlier funding with what came next. Left to right is earlier funding; bottom to top is later funding. I show the middle 90% so the dots are readable, but fit the orange line to all valid observations. Both axes use annualised rates.

Previous 1 day → next 1 day

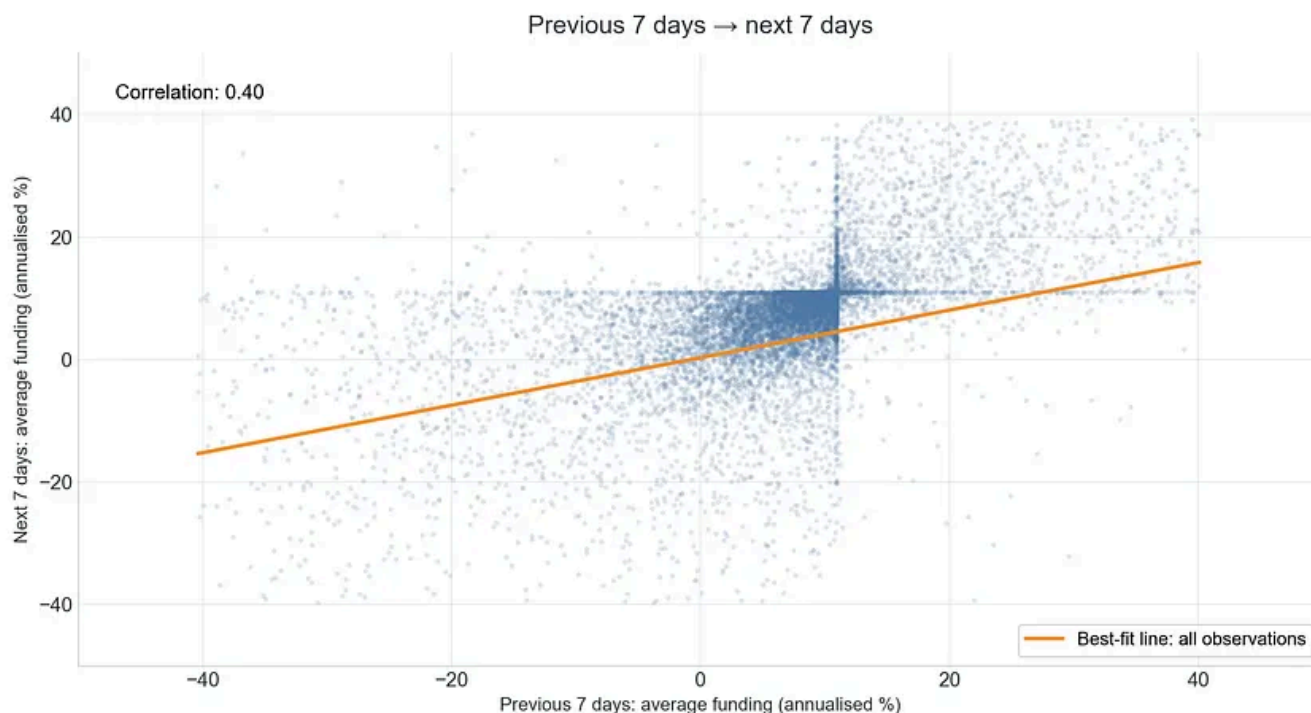


Binance archive - 2023-2025 - one dot = one coin and date - zoom: middle 90% on each axis - orange fit and correlation include the extremes outside this view.

Previous 3 days → next 3 days



Binance archive - 2023-2025 - one dot = one coin and date - zoom: middle 90% on each axis - orange fit and correlation include the extremes outside this view.



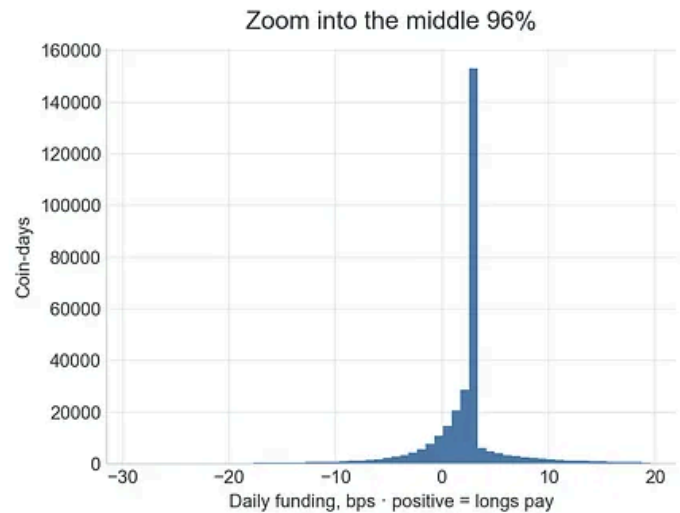
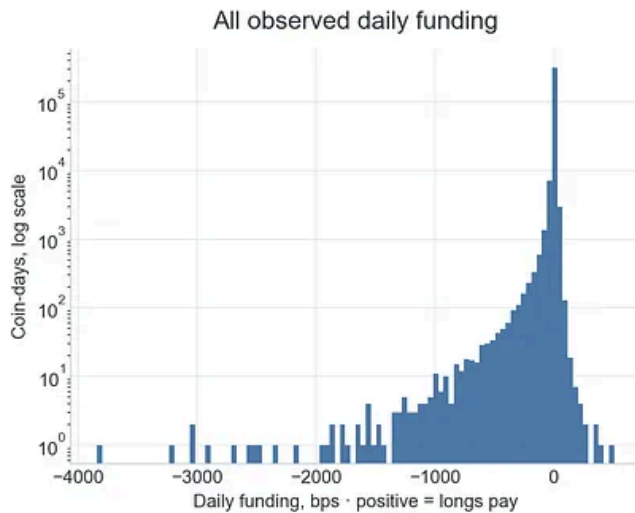
Binance archive - 2023-2025 - one dot = one coin and date - zoom: middle 90% on each axis - orange fit and correlation include the extremes outside this view.

Higher funding tended to stay higher, especially from one day to the next. That gives us a reason to keep looking. It does not yet tell us whether the payment survives the price move.

Does this depend on a few extreme rates? Removing roughly the most extreme 1% of paired observations leaves all three relationships positive. The funding pattern survives, but we still need to test returns.

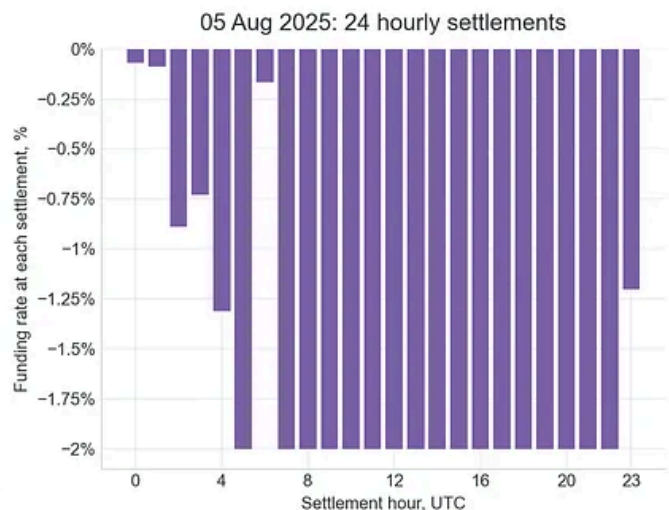
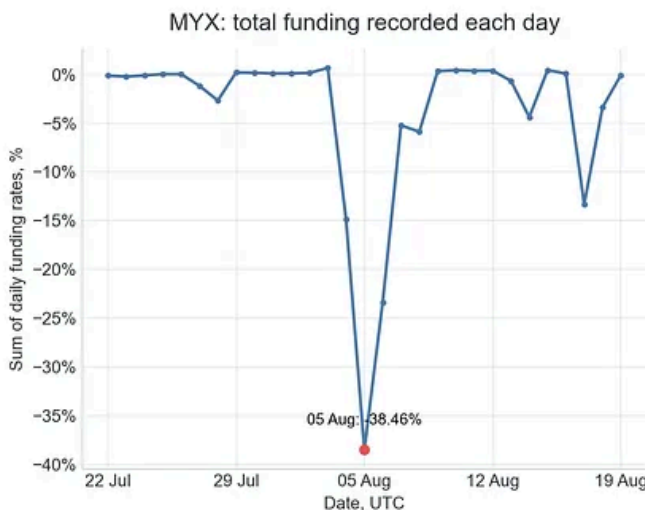
Before adding price moves, I inspect the unusual payments. Then I check how much of the funding advantage survives.

Most daily rates sit near zero, but a few are enormous. The average deserves a closer look:



Available Binance archive · 2023–2025 · complete 3-day input · observed daily payments, not annual yields · tails retained in every primary test.

The lowest daily total is -3845.8 bps (-38.46%) for MYXUSDT on 05 Aug 2025, across 24 recorded settlements. Shorts paid longs. That is a sum of rates, not a 38.46% account return: the payment depends on the position's value at each settlement. I keep it in the main tests, but it shows why these extremes need checking.



MYXUSDT · recorded Binance funding · negative = shorts pay longs · daily total is a simple sum, not a compounded return or an account return · actual payments depend on notional at each settlement.

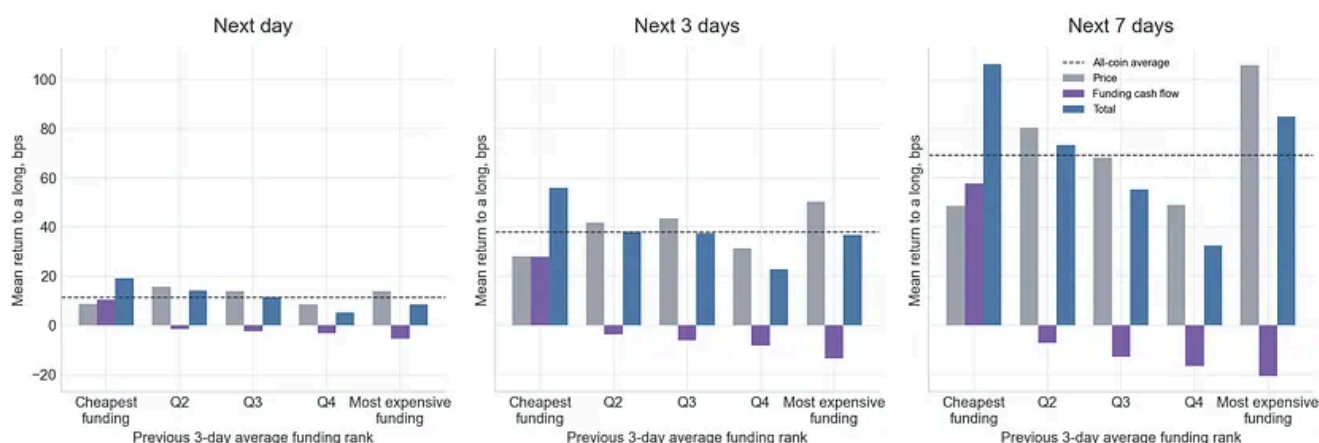
Does funding predict returns?

Now the question I actually care about: does cheaper funding leave me with more money once the price has moved?

Each day, I sort coins by their 3-day average funding and split them into five groups: Q1 is cheapest, Q5 most expensive. Tied rates stay together.

The chart separates the next 1, 3 and 7-day return into price, funding and the total. It assumes a fixed number of coins held long; a positive funding bar means we received funding. I estimate payment sizes using daily prices rather than each settlement's exact price, which matters most during big moves.

Coins get equal weight each day, then days get equal weight in the average. I keep complete outcomes rather than treating missing funding as zero. Several holds can cover the same market move, so they are not independent trades.



All contracts available in the Binance archive · 2023-2025 · 944 paired dates · fixed-quantity price + signed long-holder funding cash flow · dashed line is the all-coin average · equal-coin then equal-date means · before fees · overlapping holds.

The cheapest-funding coins had the highest average total return over the next 1, 3 and 7 days, including funding payments. Their advantage over the most expensive group averaged 10.7, 19.4 and 21.3 bps respectively, before costs. These are differences between group returns, not returns on an account.

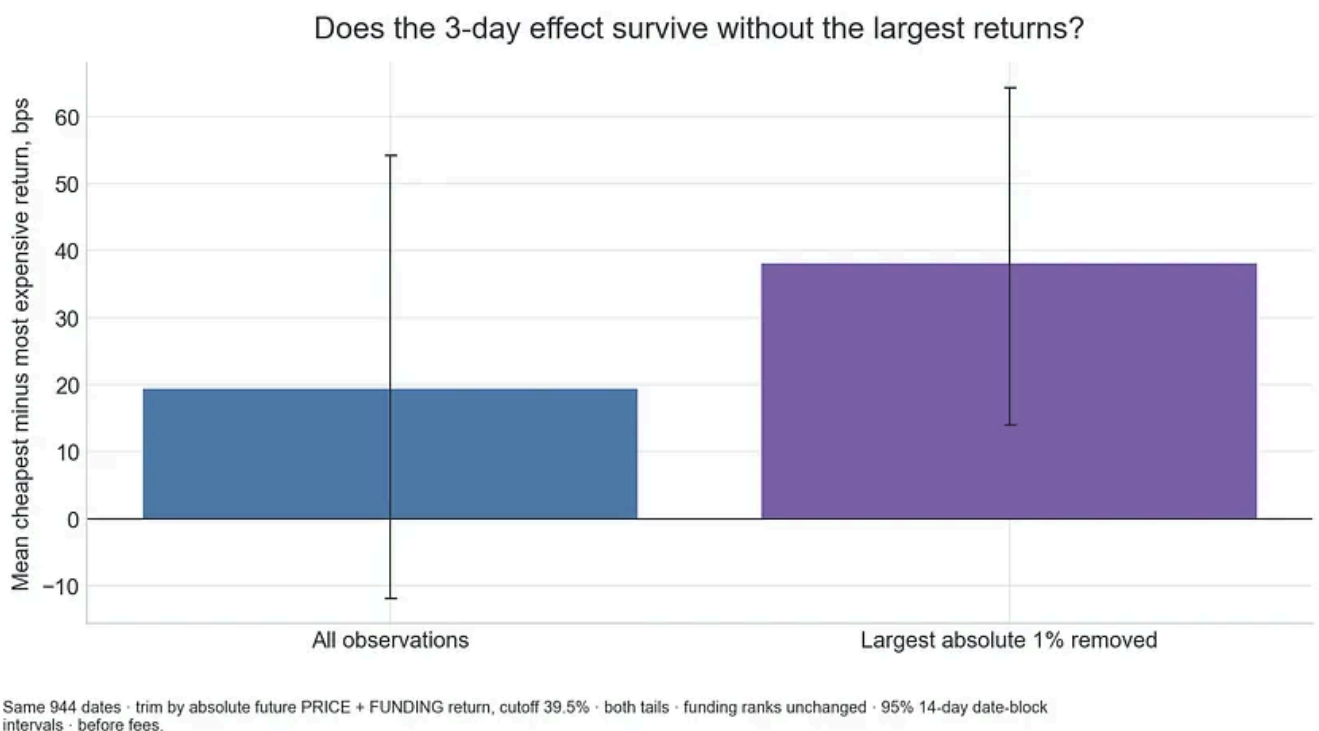
But they did not have the strongest price gains. Over 3 days, funding gave the cheapest group a 41.5 bps advantage. Weaker price performance gave back 22.2 bps, leaving 19.4 bps before costs.

The simple takeaway: some of the funding advantage survived the price moves. That is encouraging, but it varied a lot and the uncertainty ranges still include no advantage. Next I check different periods, universes and costs.

Do a few extreme moves create the average?

Crypto returns have large tails. I repeat the 3-day test after removing the largest 1% of absolute future price-plus-funding outcomes. The funding ranks do not change, and both versions use the same dates.

I cannot know tomorrow's extreme returns before trading. This is only a stress test of the average: if the result disappears without a handful of large moves, I should be more cautious about relying on it.

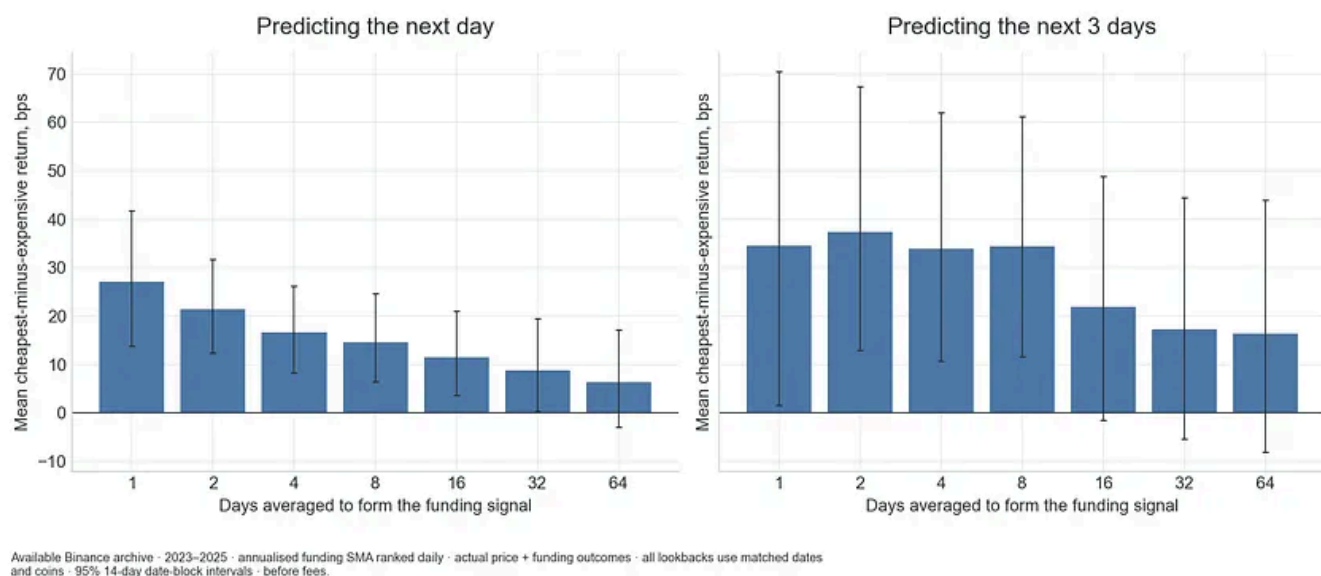


The average advantage increases from 19.4 to 38.2 bps without the largest 1% of moves. So a handful of huge winners did not create the whole pattern. The range of plausible results is still wide.

Does averaging funding for longer help?

Does yesterday's funding tell us enough, or does an average help? I try 1, 2, 4, 8, 16, 32 and 64 days. Doubling covers fast and slow signals without trying every possible number. The 3-day and 6-day rules remain the original comparisons.

For each lookback, I compare the cheapest and most expensive fifths over the next 1 and 3 days, including funding. Within each horizon, all lookbacks use the same eligible dates and coins.



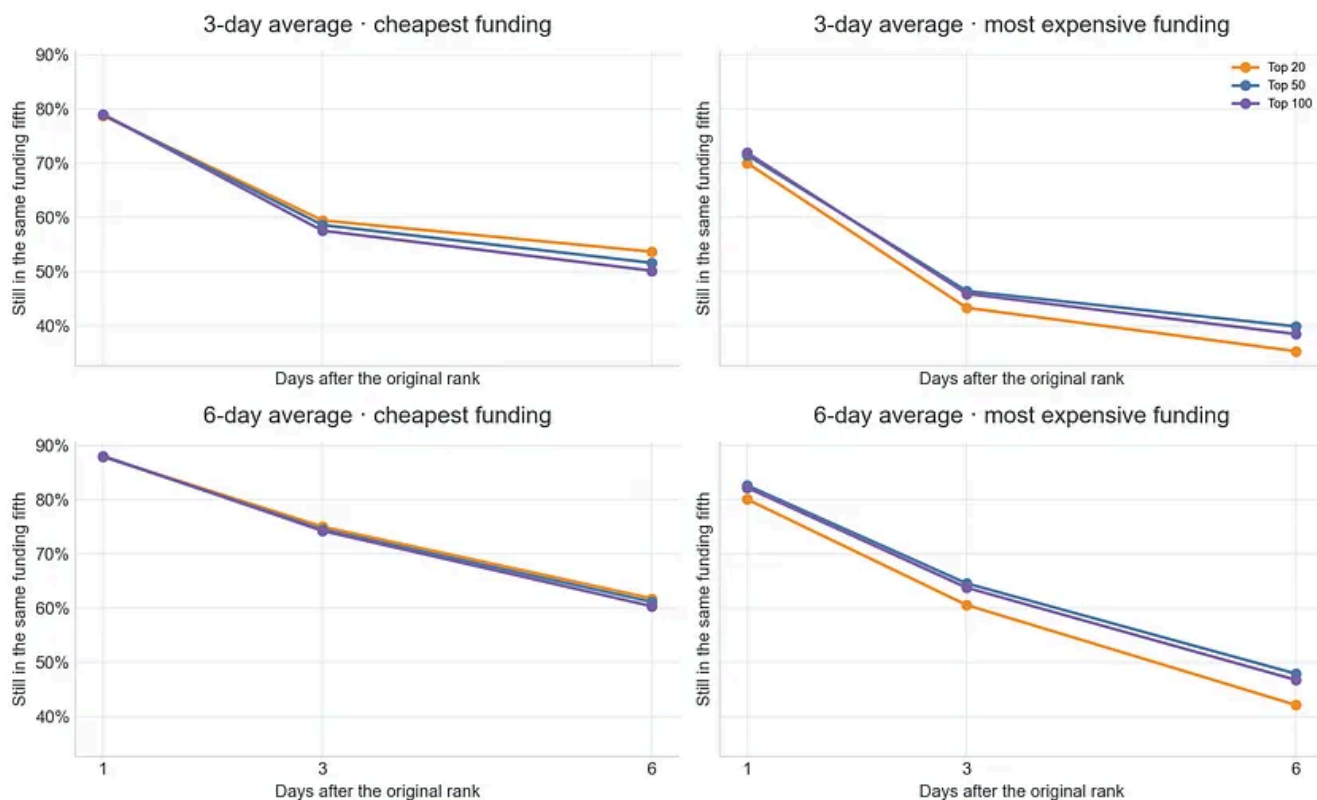
Simple takeaway: the 1–8 day averages separated cheap from expensive funding more strongly than the 32–64 day averages. Recent funding looked more useful for the next few days, but faster reactions can cost more to trade.

Do coins remain cheap or expensive?

Funding can stay high while other coins' rates move even higher. What matters for this trade is whether the same coins stay cheap or expensive. I check after 1, 3 and 6 days; a coin that leaves the universe counts as not staying.

The comparison uses 3-day and 6-day funding averages inside Top 20, Top 50 and Top 100. Stable ranks could let us trade less without abandoning the signal.

A coin staying cheap is useful only if the return covers price risk and trading costs.



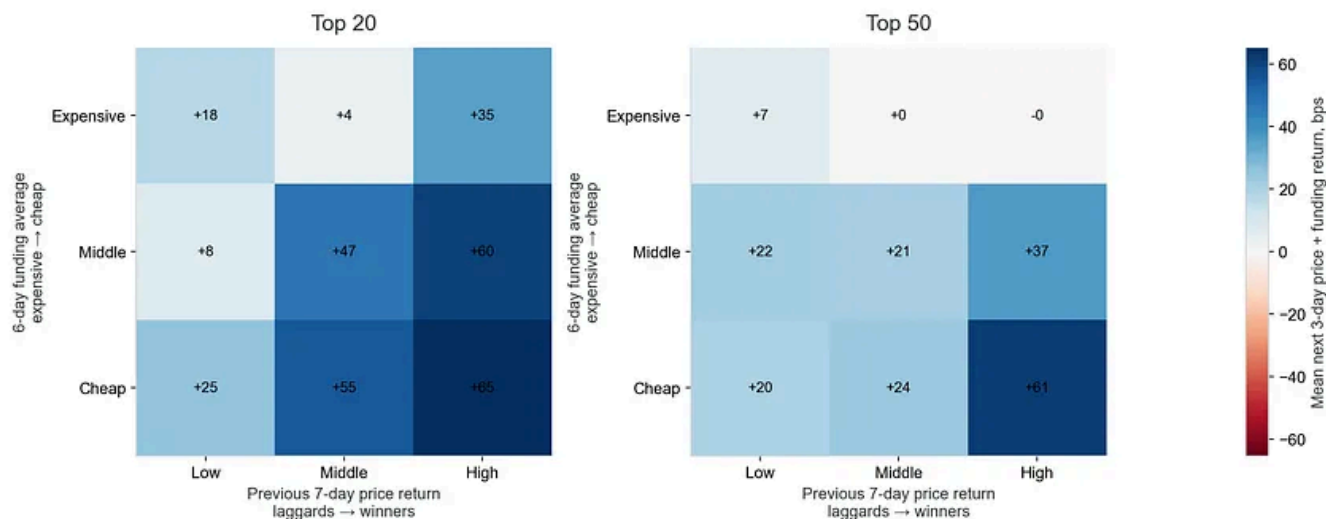
Historical market-cap Top 20, Top 50 and Top 100 · 2023–2025 · funding ranked within each universe · leaving the universe counts as not retained
 · final incomplete dates excluded · descriptive only.

With the 6-day average, about 88% of Top-20 cheap-funding observations stay cheap the next day, and 62% after 6 days. Top 50 and Top 100 look similar. That suggests we may not need to trade every day—but the returns still have to cover the bill.

Does cheap funding work better in stronger coins?

Cheap funding could just be picking coins that recently fell. To check, I separate coins into recent winners, laggards and those in between, based on their previous 7-day return. Then I compare cheap and expensive funding within each group, using the 6-day funding average.

The chart shows the next 3-day return, including funding. A “strong” coin can still be falling; it simply did better than the others. I keep only dates with all nine funding-and-momentum groups, so the result describes that smaller sample.



Historical market-cap Top 20 and Top 50 · 2023–2025 · independent daily thirds of prior 6-day funding and 7-day price return · only dates with all nine cells · next 3-day fixed-quantity total return · before trading costs.

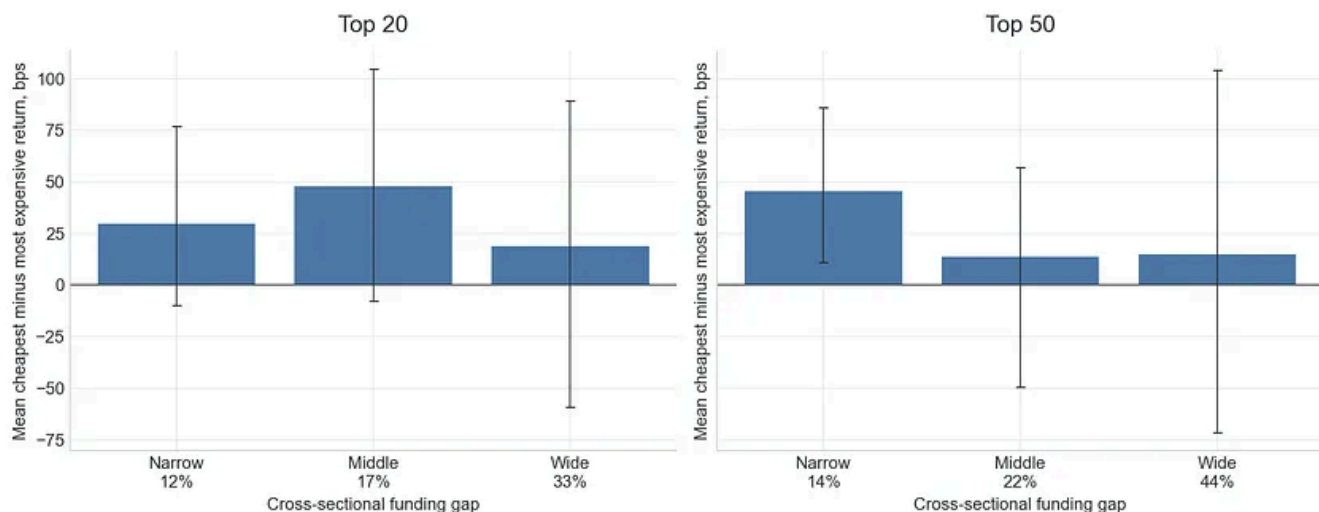
Cheap funding beat expensive funding in all three momentum groups. The strongest average returns came from coins that combined cheap funding with the strongest recent price performance. That pattern appeared in both Top 20 and Top 50.

The simple takeaway: cheap funding looked most promising alongside relative price strength. That is a useful combination to investigate, not yet a tested combined strategy after costs.

Does a wider funding gap contain more information?

A relative rank shows order, not distance: 1% versus 2% annualised funding is a smaller gap than 1% versus 30%. I test whether that gap tells us when carry is more useful.

I group retained dates by the annualised funding gap between the expensive and cheap fifths, then measure their following 3-day return spread.



Historical market-cap Top 20 and Top 50 · 2023–2025 · previous 6-day funding average · dates split into equal thirds by annualised most-expensive-minus-cheapest funding gap · next 3-day price + funding · 95% 14-day date-block intervals · before costs.

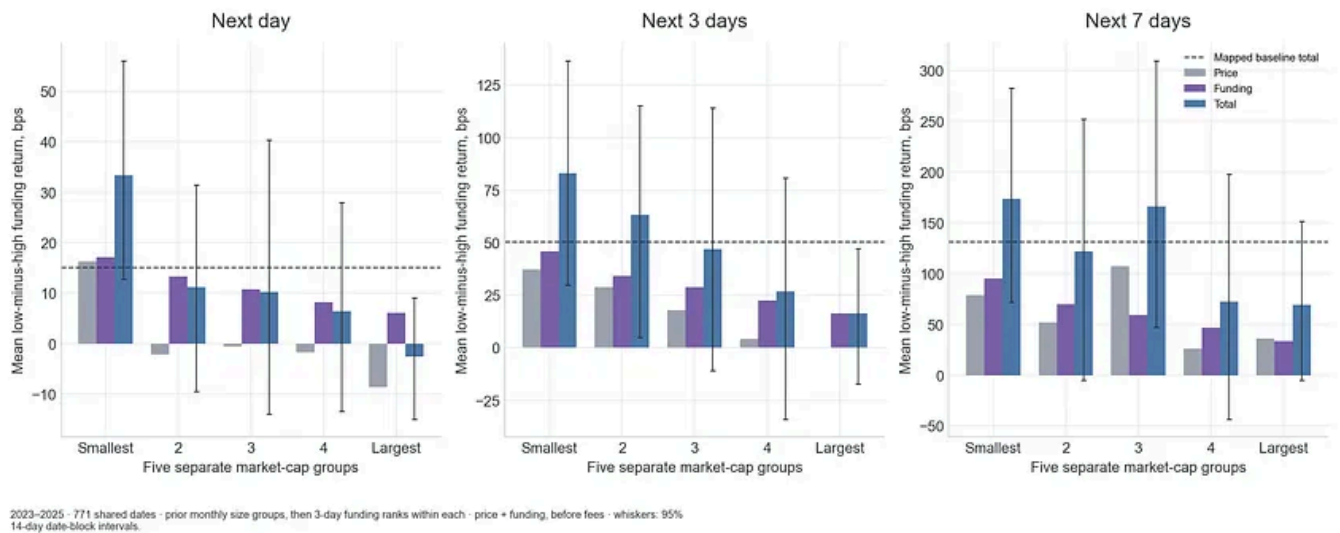
A bigger funding gap did not reliably mean a bigger return. I would not use “the gap looks unusually wide” as a reason to trade more from this evidence. The percentages below the bars describe average annualised funding gaps, not live trading thresholds. Most uncertainty intervals include zero.

Does the effect depend on market cap?

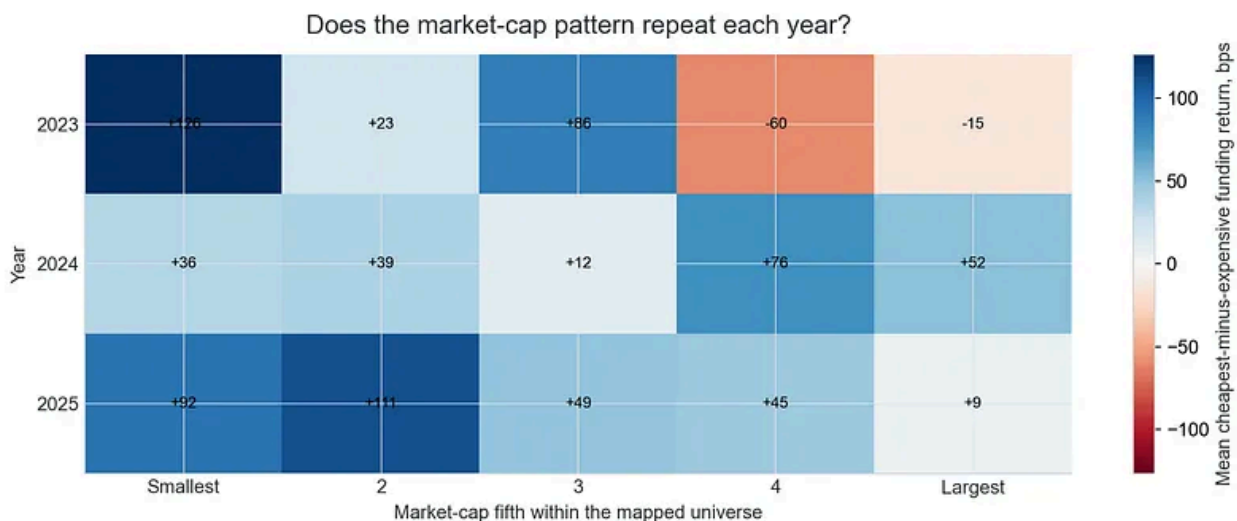
I split the mapped contracts into five equal market-cap groups each month, using a snapshot known before that month. Then I rank funding again inside each size group. This asks whether cheap funding beats expensive funding among coins of roughly similar size.

There are two different sorts here: first large versus small coins, then cheap versus expensive funding *within* each size group. A cheap-funding coin is not necessarily a small coin.

These are size groups within our mapped Top-200 coverage, not every Binance coin. The bars compare cheap with expensive funding inside each group. The lines around them show uncertainty: longer lines mean less confidence in the average.



The smallest covered coins show the largest spread at each horizon. But the intervals are wide and several overlap. That makes the smaller coins interesting, not automatically better. They are also likely to cost more to trade.



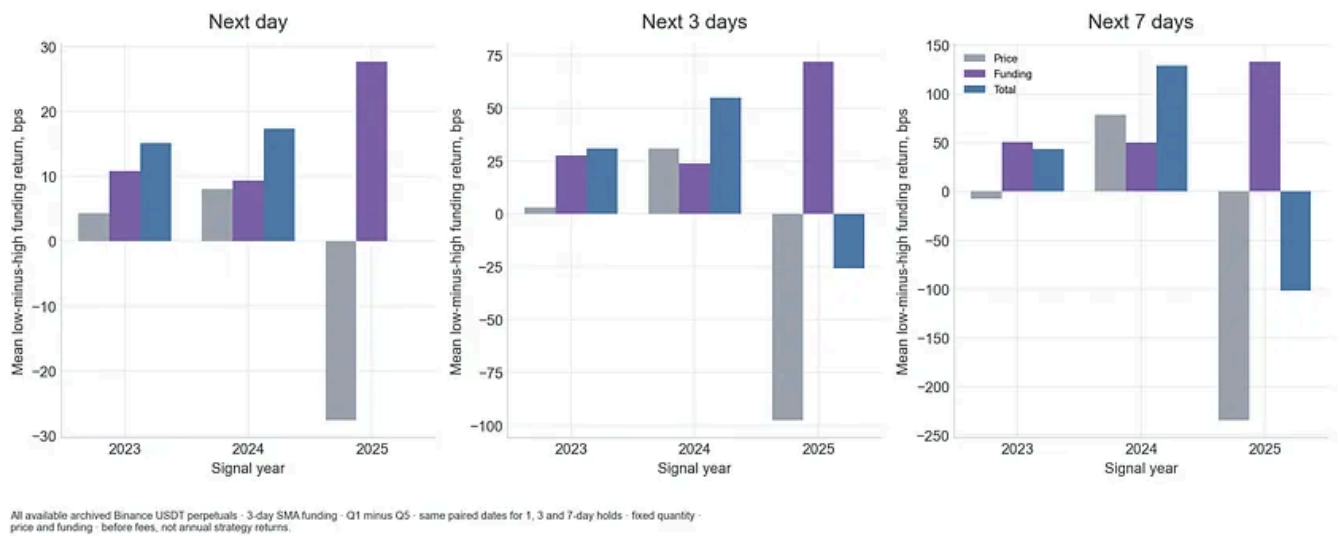
Historical market-cap snapshots known before each month · 2023–2025 · 3-day funding average ranked within each size fifth · price + funding · before fees · descriptive yearly means.

The smaller groups look stronger overall, but not consistently every year. A bigger before-cost difference is interesting only if it survives the extra difficulty of trading those coins.

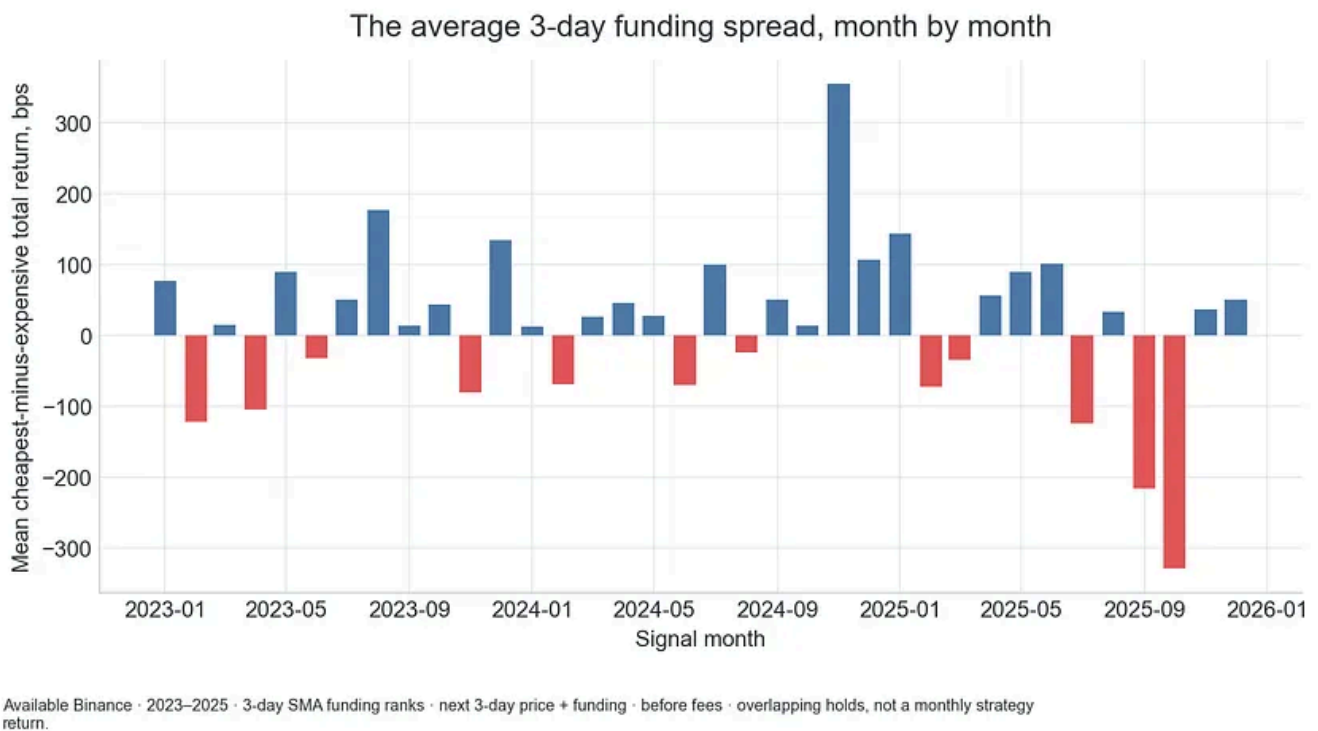
Does the broad result survive different years?

Did one good year create the average? I split the broad-universe result by year and the 3-day spread by month.

I use this to check consistency, not to choose when to trade.



The 3-day spread was positive in 2023 and 2024, then negative in 2025. In 2025, the more favourable funding did not compensate for the price move. Holding longer did not fix that year.



There are only three full years here. Splitting them into months helps me see where the average came from; it does not give me more years of evidence.

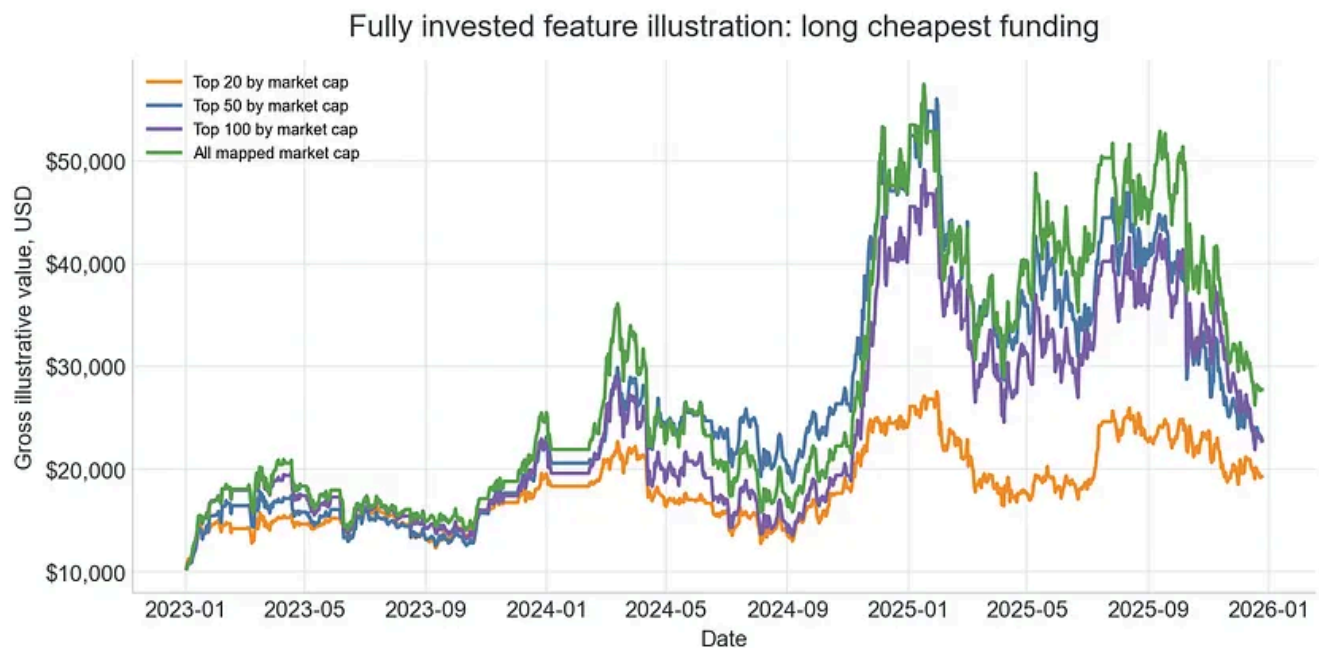
What do the cheapest and most expensive groups look like?

The bar charts show average observations. A cumulative line makes the path easier to understand. First I test the obvious long-only question: buy the cheapest funding fifth inside the mapped Top 20, Top 50, Top 100 or entire mapped universe. The groups are rebuilt each day using the 3-day funding average.

The Top-50 chart shows the long-cheap leg, the short-expensive leg and their 50/50 combination. The purple line is simple cross-sectional carry: equal dollars long and short.

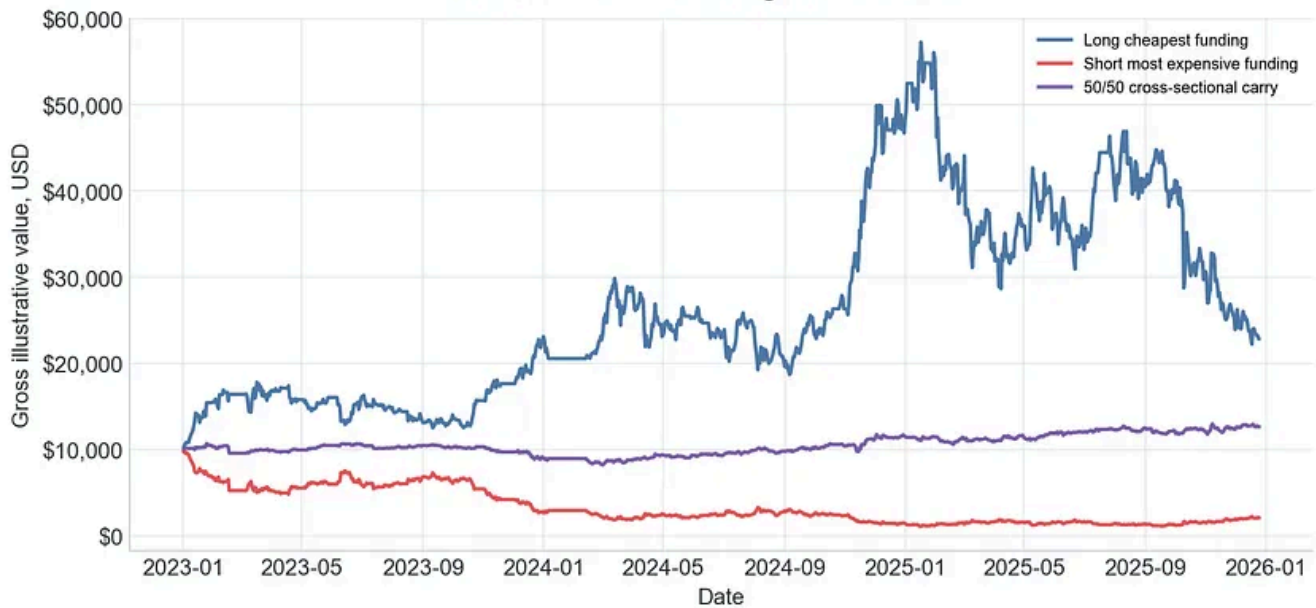
These illustrations are fully invested, equal-weighted within each group and rebalanced daily, before costs. They are not the capped portfolios in [Part 2](#).

Splitting the legs answers a useful question: does the short side actually help, or am I attaching a losing short trade to a decent long trade?



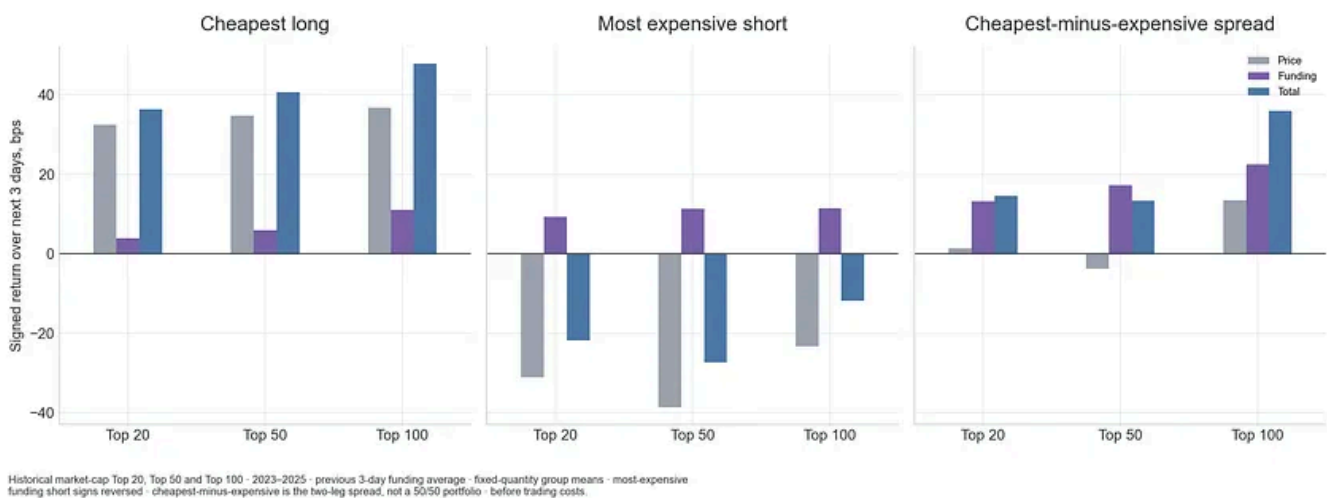
\$10,000 start · feature illustration, not the later strategy · 2023–2025 · historical market-cap snapshots known before each month · daily 3-day funding groups · 100% gross equal weight within the cheapest funding fifth · price + funding · before trading costs.

What did the two funding tails contribute?



\$10,000 start · mapped market-cap Top 50 · 2023–2025 · daily 3-day funding groups · equal weight within the cheapest and most expensive funding fifths · price + funding · before trading costs · 50/50 spread has 100% gross.

Where does the universe difference come from?



Historical market-cap Top 20, Top 50 and Top 100 · 2023–2025 · previous 3-day funding average · fixed-quantity group means · most-expensive funding short signs reversed · cheapest-minus-expensive is the two-leg spread, not a 50/50 portfolio · before trading costs.

This is why I split the two sides: buying cheap funding and shorting expensive funding are not equally attractive. A decent long trade can be dragged down by a losing short. The chart compares the next 3 days.

The cheapest funding fifth contains 4.0 coins on average inside Top 20 and 10.0 inside Top 50. This is a funding group formed inside the market-cap universe; it is not the smallest market-cap group.

What survives costs in the simplest long/short rule?

Before changing the sizing, I keep the simple rule: rank the historical Top 50 by their last 3 days of funding, put half the account long the cheapest fifth and half short the most expensive fifth. Coins have equal weights within each side.

Check the positions daily.

I compare the same positions before and after trading costs: 10 bps, or \$10 for every \$10,000 traded. Entry, resizing and the final exit all count. There is no volatility sizing, buffer or extra smoothing. Half the account is long and half short when five groups can be formed.

The portfolio runs through every day in 2023–2025, holding cash when five funding groups cannot form: 985 active days and 110 cash days. Missing returns on held coins stop the test. Ties can leave very few coins in a group, so the book can be concentrated. This wider calendar also explains why the gross line differs from the earlier illustration.



\$10,000 start · historical market-cap Top 50 · 2023–2025 · 3-day funding SMA · 50% long / 50% short, equal coin weights · daily rebalance · price + funding · 10 bps one-way, entry and final exit included · no volatility sizing or 2.5% cap · cash if five groups unavailable.

The simple rule turns \$10,000 into \$12,740 before costs, but \$7,470 after costs. Gross sr is 0.55; net sr is -0.47. Annual turnover is 177.9× and net max dd is

-32.7%. The trading bill ate the return. Here sr compares average return with volatility, and max dd is the largest peak-to-trough fall.

What I take from this

Funding contained an interesting pattern in this sample. Cheap-funding coins had better average total returns than expensive-funding coins, but price moves took back part of the payment advantage. The result also changed with the year and the size of the coins.

The useful rules of thumb are:

- Count the price move as well as the funding payment. Getting paid does not mean making money.
- Recent funding looked more useful than a long average for the next few days.
- Check the long and short sides separately. They did not contribute equally.

The simplest daily long/short rule lost money after costs.

In [Part 2](#), the question is practical: can we keep enough of that return by trading less? I start with the same rule and change the review schedule before trying different position sizes.

Discuss or compare notes

I'm a self-taught systematic trader. I will make mistakes and correct them. If you spot an error or have a better test, message me on [X: @dima_quant](#).

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